SOFTWARE ENGINEERING LAB – 1

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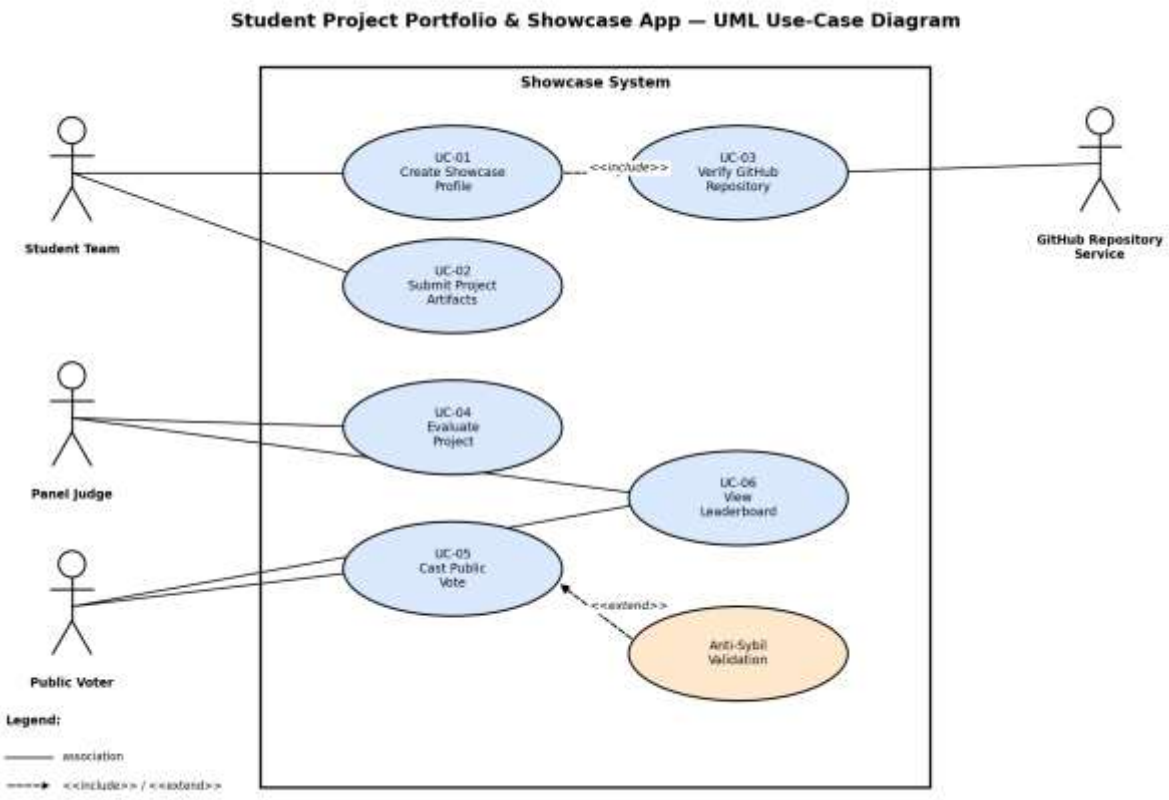
SRN: PES2UG24CS372

SECTION: G

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable student teams to create showcase profiles containing abstract summaries, demo video links, and verified GitHub repository URLs.	High	**Pass:** Profile is published to the showcase catalog after the repository URL is successfully validated. **Fail:** An invalid repository URL is accepted or the profile is published without validation.	Core functionality for presenting student projects in the showcase.
FR-002	Functional	The system shall allow student teams to submit and manage multimedia project artifacts for their showcase profile.	High	**Pass:** A team can add supported project artifacts and they are displayed in its project profile. **Fail:** An unsupported artifact is accepted or an artifact is associated with the wrong project.	Enables teams to present projects using multimedia content.
FR-003	Functional	The system shall allow panel judges to evaluate submitted projects using a digital scoring rubric.	High	**Pass:** A judge can enter scores for every required rubric criterion and submit the evaluation. **Fail:** Submission is allowed with a required criterion missing.	Provides the required digital judging functionality.
FR-004	Functional	The system shall allow eligible public participants to cast a vote for a project while enforcing anti-sybil voting controls.	High	**Pass:** An eligible participant can cast one valid vote according to the configured rules, while a repeated or invalid attempt is rejected. **Fail:** Prohibited repeated voting succeeds.	Supports public participation while protecting voting integrity.
FR-005	Functional	The system shall calculate and display project rankings based on the configured judge scores and valid public votes.	High	**Pass:** After a valid score or vote is recorded, ranking data is recalculated and displayed correctly. **Fail:** Invalid votes or scores affect the leaderboard.	Provides a centralized and transparent showcase leaderboard.
NFR-001	Nonfunctional — Performance & Security	The public voting and leaderboard calculations shall update in real-time with an API response latency under 300 ms.	High	**Pass:** Benchmarking tests confirm the target latency and security standards under simulated peak load. **Fail:** API response exceeds 300 ms under the defined test conditions.	Ensures responsive voting and leaderboard updates during peak participation.
NFR-002	Nonfunctional — Security	The system shall protect public voting against unauthorized or duplicate voting attempts using the configured anti-sybil controls.	High	**Pass:** Security testing demonstrates that unauthorized and duplicate voting attempts are rejected according to configured rules. **Fail:** A prohibited vote bypasses the controls.	Maintains fairness and integrity of public voting.

UML Use-Case Diagram



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Use-Case Flow Specification Use Case: UC-04 — Evaluate Project (Student Project Portfolio & Showcase App)

Preconditions

- The project has been successfully submitted by a student team.
- The project is available for judging.
- The panel judge has access to the judging functionality.
- A digital scoring rubric is available for the project.

Postconditions

- The judge's scores are stored successfully.
- The evaluation is associated with the correct project.
- The evaluation can be used in the project's ranking/leaderboard calculation.

Main Success Scenario

1. The Panel Judge selects a project from the showcase.

2. The system displays the project's showcase profile and submitted artifacts.

3. The Panel Judge selects **Evaluate Project**.

4. The system displays the configured digital scoring rubric.

5. The Panel Judge enters a score for each required rubric criterion.

6. The system validates the entered scores.

7. The Panel Judge submits the completed evaluation.

8. The system stores the evaluation against the selected project.

9. The system confirms that the evaluation has been submitted successfully.

10. The system updates the project's ranking data using the newly recorded judge score.

11. The use case ends successfully.

Alternate Flow — Invalid/Missing Score

1. A required rubric score is missing or invalid.
2. The system detects that one or more rubric criteria do not contain valid scores.
3. The system displays an error identifying the incomplete or invalid criterion.
4. The Panel Judge is prompted to correct the missing or invalid score.
5. The Panel Judge enters valid values.
6. The system validates the scores again.
7. If all required scores are valid, the flow continues at Step 7 of the Main Success Scenario.